engineers, UAV pilots and test facilities leaves prototypes untested for ruggedness or electronic warfare resilience," says a defence analyst. Many Indian drones, he adds, rely on cheap parts that fail in extreme environments, making them vulnerable to jamming.

While government initiatives such as the Production Linked Incentive (PLI) scheme and iDEX (Innovation of Defence Excellence) provide support, they are deemed insufficient for advanced R&D. "Import dependency, policy inertia, limited talent and high production costs constrain India's drone ambitions," says an industry insider.

"The military must first articulate precise requirements, specifying the indigenous components they seek in drones," says Group Captain R.K. Narang, senior fellow at the Manohar Parrikar Institute for Defence Studies and Analyses. "They should allow the industry sufficient time to develop these technologies."

According to Major General C. S. Mann, additional director general of the Army Design Bureau (ADB), the challenges that need to be overcome include enemy interception of communication links and hijacking of drones' navigation control. "The components used in flight controllers, transmitters and receiver units, encryption and authentication systems can have vulnerabilities," he says. Maj. Gen. Mann is the architect of a framework intended to secure India's military drone ecosystem from security-compromising foreign intrusions. It will redefine the procurement process of UAVs, but has yet to be approved by the defence ministry.

Lessons and Gaps

One of the key lessons learnt during Op. Sindoor was the jamming of India-made drones. Pakistani forces employed high-energy blanket GPS jamming across a 150 km radius, disrupting L-band signals and hampering drone activity. In response, Indian forces raised drone altitudes to 1.5 km, but the enemy countered with directional jamming, targeting drones with antenna-based interference. With secure satellite communication indispensable for maintaining encrypted, uninterrupted links between UAVs and control stations, UAVs must incorporate stealth and electronic countermeasure features.

Two months after Op. Sindoor, the Indian army invited 46 Indian drone firms to showcase their capability to perform in a GPS-denied environment in Uttarakhand. None passed the trials.

Indian firms aren't that interested in advanced R&D, admits a drone manufacturer. Also, the 'no cost, no commitment' trials—where producers develop products at their own expense, with no guarantee of purchase—conducted by the military are an expense firms can ill-afford. "Once a drone-maker completes a trial conducted by a military unit, why does he have to compete again for a fresh tender? It costs too much," says Sai Pattabiram, MD of Chennai-based Zuppa Geo Navigation Technologies, manufacturer of the Hawk and Scout drones for the Indian military.

India began working on drones in the 1980s, around the same time as countries like Israel. However, the state-owned Defence Research and Development Organisation (DRDO), which took up the mission, doesn't still have a single UAV in the military's inventory. Its ISR drone projects—Lakshay, Nishant, Rustom and Tapas—haven't gone anywhere. India, to be blunt, faces a massive technological and capability gap.

A senior DRDO scientist says that global advancements in sensor payloads such as electro-optical and intelligence systems—critical to surveillance and data collection—have left them behind. "Modern UAVs demand advanced radars and synthetic aperture variants capable of providing high-resolution imaging," he says. Another pressing challenge, he adds, is the need for robust multi-sensor data fusion to enable real-time situational awareness. Propulsion technologies also require innovation—advancements in internal combustion and jet engine systems are essential to increase range, endurance and altitude of drones.

Op. Sindoor highlighted the urgent need to both improve the quality of drones and to develop robust counter-drone technologies. The Indian military and drone-makers must take up the challenge to stay ahead in this rapidly evolving domain. ■

ROADBLOCKS TO SURMOUNT

The challenges before the Indian drone industry faces

SUSCEPTIBILITY TO ELECTRONIC WARFARE: Many Indian drones fell prey to hijacking, spoofing and jamming by Pakistani forces during Op. Sindoor; Israeli drones unaffected

LACK OF SWARM CAPABILITY Indigenous UAVs, even from the same firm, were unable to coordinate with each other effectively in swarm attacks

POOR TEST RESULTS: Post the conflict, no Indian drone passed the GPS-denied environment trials in Uttarakhand

OVER-RELIANCE ON FOREIGN PARTS Around 70% of critical drone components—motors, sensors and flight controllers—are imported, mostly from China, a marker of India's technical/capability gap

DELAYED UAVs Despite generous government grants, indigenous military drones to be developed by DRDO—Nishant, Rustom and Tapas—are still not fielded

After Op. Sindoor, the army asked 46 Indian drone-makers to showcase their ability to perform in a GPS-denied environment in Uttarakhand. None passed the trials